

ICT: Business intelligence & CRM Databases

Anna Monreale

Università di Pisa

General Information

- **Anna Monreale**

- Email: anna.monreale@unipi.it

- **Learning Material**

- Slides
 - Databases essentials – Antonio Albano (pdf)
 - A First Course in Database Systems, Jeffery D. Ullman and Jennifer Widom

Topics

- **Databases**
 - What is a DBMS
 - Designing a DB
 - Querying a DB

Information as a Resource

- **Information** is one of the most important resources of any **organization**
- An intelligent management of the information can help organization to generate **new knowledge**
- It becomes more and more important learning how to represent, organize, manage and use information

Organization

- **Organizations**
 - es: companies, banks, public administrations,
- An organic collection of **resources** (people, materials, **information**), **tools** and **procedures**, which are finalized to create and offer a product or a service
 - a bank provides financial services
 - a hospital supplies medical services

Information System

- Component of an organization finalized to the **information management** for supporting the organization activities
 - Collects, stores, processes and communicates the information
- Any organization has an Information System

Information System & Computerization

- **Information system** is independent on the computerization
- Organizations as Libraries, Banks, etc ... existed before the computerization and managed information

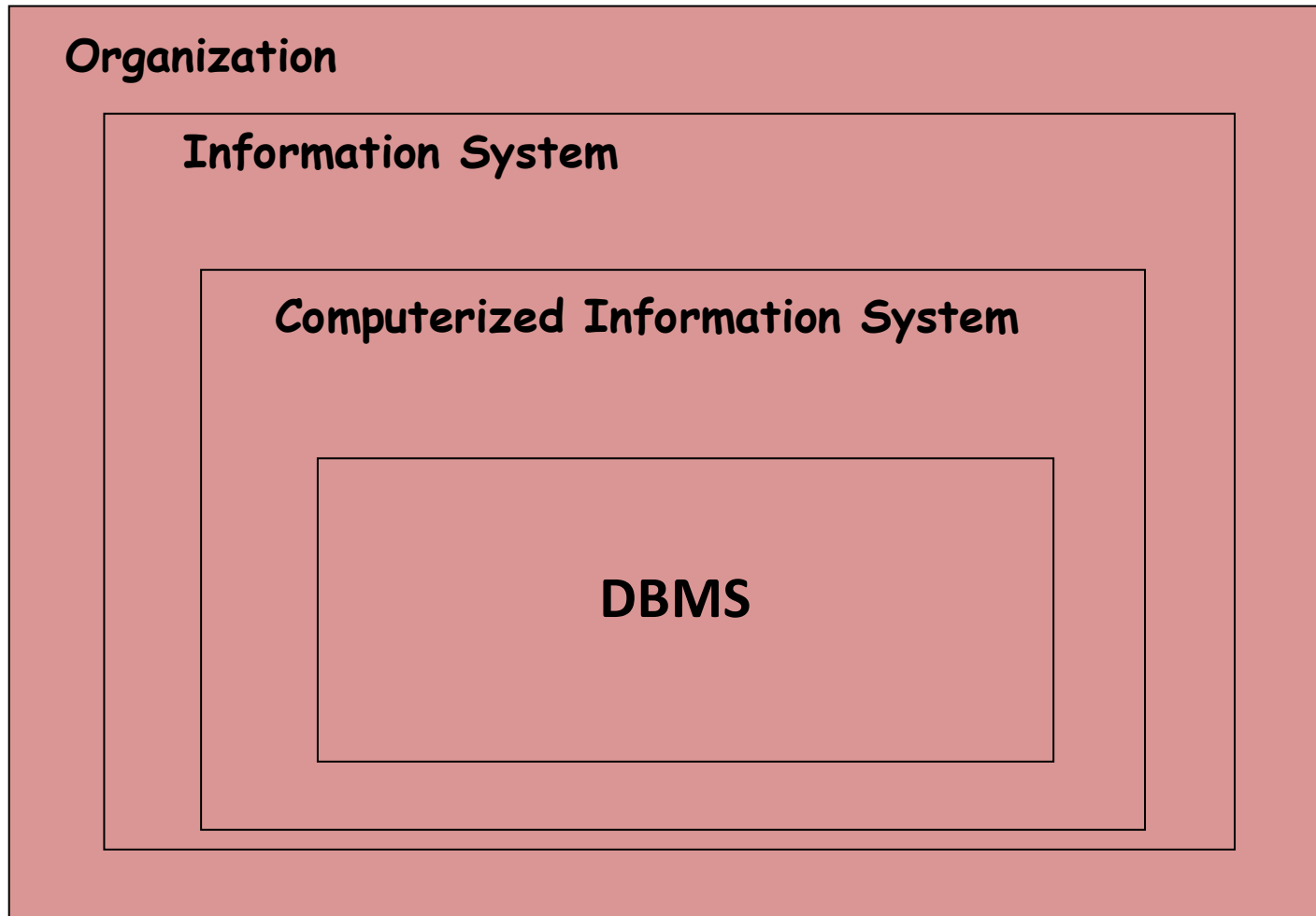


Real Library of Alexandria d'Egypt

Computerized Information System

- The hardware and software used for storing, retrieving, and processing the information which supports the functions of an organization
- In practice, in many cases:
 - **information system** is used as synonym of **computerized information system**
 - But it is not always true
- Implemented by **database technology**
 - operational database and a collection of application programs used to access and update the data quickly and efficiently

Computerized Information System

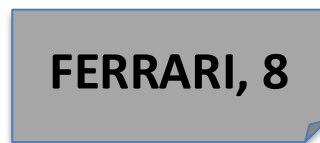


Goal of DBMS: processing data for getting information

Data vs Information

- In information systems **information** is represented by **data**
- **Information** (def): news, or element enabling knowledge about facts and situations
- **Datum** (def): element of an information composed of symbols to be processed

Without interpretation datum is useless



Database

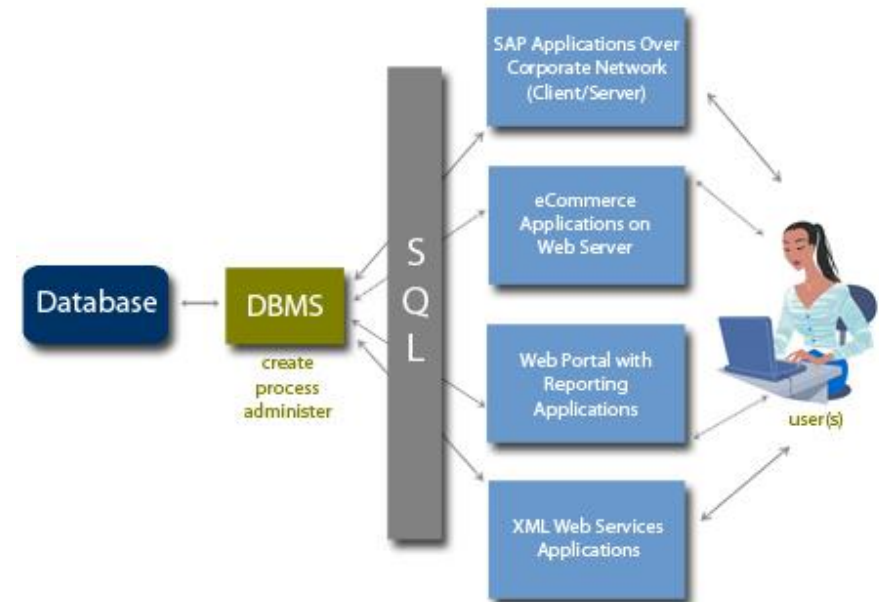
- Collection of permanent data representing facts interesting for an organization (*data or instances*) and facts related to the data organization (*metadata or schema*)

STUDENTS

ID	Surname	Name	BirthofDate
276545	Rossi	Maria	25/11/1991
485745	Neri	Anna	23/04/1992
200768	Verdi	Fabio	12/02/1992
587614	Rossi	Luca	10/10/1991
937653	Bruni	Mario	01/12/1991

Database Management Systems

- A set of tools, to manage homogeneous sets of structured data
- Able to deal with
 - Huge amounts of data
 - Mission critical data
 - Shared data
 - Queries and updates



Some DBMS

- **Commercial**
 - IBM DB2, Oracle, **Microsoft SQL Server**, Sybase
 - Microsoft Access, FileMaker
- **Open Source**
 - MySQL (www.mysql.com)
 - PostgreSQL (www.postgresql.org)

Databases

Relational Model

Elements of the model

- **Database**
 - Set of **tables**
- **Table**
 - Set of **records**
- **Record**
 - Set of pairs (**attribute, value**)

Example: Students, Courses, Exams

- **Database: University**
- **Students**
 - name, surname, studentID, date of birth
- **Courses**
 - code, title of course, teacher
- **Exams**
 - course, student, mark

Example: Students, Courses, Exams

- **Students**

- **name**: string
- **surname**: string
- **studentID**: integer
- **date of birth**: date

- **Courses**

- **code**: string
- **title**: string
- **teacher**: string

- **Exams**

- **course**: “reference” to a course
- **student** “reference” to a student
- **mark**: integer
- **laud**: yes/no - boolean

Students

Table: relation
(instance)

record

Attribute: property

value

Students	StudentID	Surname	Name	Date of Birth
	276545	Rossi	Maria	25/11/1991
	485745	Neri	Anna	23/04/1992
	200768	Verdi	Fabio	12/02/1992
	587614	Rossi	Luca	10/10/1991
	937653	Bruni	Mario	01/12/1991

Relational Schema

```
TABLE Students(  
  studentID integer,  
  surname char(20),  
  name char(20),  
  DateofBirth date);
```

**Domain
Type**

Attributes

Students	StudentID	Surname	Name	Date of Birth
	276545	Rossi	Maria	25/11/1991
	485745	Neri	Anna	23/04/1992
	200768	Verdi	Fabio	12/02/1992
	587614	Rossi	Luca	10/10/1991
	937653	Bruni	Mario	01/12/1991

```
TABLE Students(  
    studentID integer,  
    surname char(20),  
    name char(20),  
    DateofBirth date);
```

- Each attribute has a **domain** defining the **set of valid values** for the attribute
Ex. $\text{dom}(\text{studentID}) = \text{integer}$
- We can have a repetition of domains in the same relation!

Constraints on the order of the data

Students



StudentID	Surname	Name	Date of Birth
276545	Rossi	Maria	25/11/1991
485745	Neri	Anna	23/04/1992
200768	Verdi	Fabio	12/02/1992
587614	Rossi	Luca	10/10/1991
937653	Bruni	Mario	01/12/1991

- Order of records is not important
- Order of columns is not important

Constraints on data

Students



StudentID	Surname	Name	Date of Birth
20/11/1991	Rossi	Maria	25/11/1991
485745	Neri	Anna	23/04/1992
200768	Verdi	Fabio	12/02/1992
587614	Rossi	Luca	10/10/1991
937653	Bruni	Mario	01/12/1991



- No equal **attributes** (1)
- No **equal rows** (2)
- Data of **column** must be **homogeneous** (3)

Courses

COURSES

Code	Title	Teacher
01	Analisi	Giani
03	Chimica	Melli
04	Chimica	Belli

```
TABLE Courses (  
  code char(2) ,  
  title char(50) ,  
  teacher char(20)) ;
```

Exams

Exams

Student	Mark	Laud	Course
276545	28	0	01
276545	27	0	04
937653	25	0	01
200768	30	1	04

StudentID
of a
student



Code of a
course



```
TABLE Exams (  
    student integer,  
    mark integer,  
    laud bool,  
    course char(3) );
```

Tables

Students

StudentID	Surname	Name	Date of Birth
276545	Rossi	Maria	25/11/1991
485745	Neri	Anna	23/04/1992
200768	Verdi	Fabio	12/02/1992
587614	Rossi	Luca	10/10/1991
937653	Bruni	Mario	01/12/1991

Exams

Student	Mark	Laud	Course
276545	28	0	01
276545	27	0	04
937653	25	0	01
200768	30	1	04

Courses

Code	Title	Teacher
01	Analisi	Giani
03	Chimica	Melli
04	Chimica	Belli

DB Schema

```
TABLE Students(  
    studentID integer,  
    surname char(20),  
    name char(20),  
    DateofBirth date);
```

```
TABLE Exams(  
    student integer,  
    mark integer,  
    laud bool,  
    course char(3));
```

```
TABLE Courses(  
    code char(3),  
    title char(50),  
    teacher char(20));
```

NULL Values

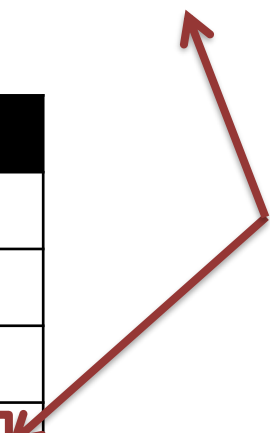
Students

StudentID	Surname	Name	Date of Birth
276545	Rossi	Maria	25/11/1991
485745	Neri	Anna	23/04/1992
200768	Verdi	Fabio	12/02/1992
587614	Rossi	Luca	10/10/1991
937653	Bruni	Mario	01/12/1991
993354	Gialli	Lucia	null

Courses

Code	Title	Teacher
01	Analisi	Giani
03	Chimica	Melli
04	Chimica	Belli
05	Basi Dati	null

Null Value



Constraints on Data

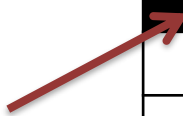
- **Rules of the scenario**
- **Unicity of identifiers (keys)**
 - Code of courses and studentID
- **Conditions on values of the records**
 - Students mark
 - Range: 18 - 30
 - Laud only if mark is 30
- **Correctness of references**

Primary Key

- **Key**: a minimal set of attributes **uniquely identifying** tuples in a relation

Students

Key		StudentID	Surname	Name	Date of Birth
		276545	Rossi	Maria	25/11/1991
		485745	Neri	Anna	23/04/1992
		200768	Verdi	Fabio	12/02/1992
		587614	Rossi	Luca	10/10/1991
		937653	Bruni	Mario	01/12/1991



- {*Surname, Name*}: is not a real Key!
- A (Primary) Key cannot have *null value*

Database with errors

Students

StudentID	Surname	Name	Date of Birth
276545	Rossi	Maria	25/11/1991
485745	Neri	Anna	23/04/1992
200768	Verdi	Fabio	12/02/1992
937653	Rossi	Luca	10/10/1991
937653	Bruni	Mario	01/12/1991

Unicity



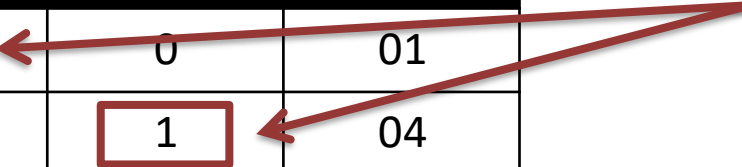
Exams

Student	Mark	Laud	Course
276545	32	0	01
276545	27	1	04
937653	25	0	01
300300	30	1	04

Reference



incorrect
mark



Integrity Constraints

- Rules on data
- Constraints on single tables
 - **Key Constraints**
 - **Constraints on tuples**
- Constraints between tables
 - **Constraints of referential integrity**

Integrity Constraints

- **Key Constraints**
- Key: identifier for tuples
 - ex: “studentID” is a Key for “Students”
- **Constraints on tuple**
 - Predicates on the values of tuples
 - ex: (mark $\geq$ 18 and mark $\leq$ 30)
- **Referential Constraints**
 - Absence of references that do not exist
 - ex: exist Exams only for Students of BD

Key Constraints

Students	StudentID	Surname	Name	Date of Birth
	276545	Rossi	Maria	25/11/1991
	485745	Neri	Anna	23/04/1992
	200768	Verdi	Fabio	12/02/1992
	587614	Rossi	Luca	10/10/1991
	937653	Bruni	Mario	01/12/1991

```
TABLE Students (  
  studentID integer PRIMARY KEY,  
  FiscalCode char(16),  
  surname char(20),  
  name char(20),  
  DateofBirth date,  
  UNIQUE (FiscalCode) );
```


Constraints on tuples

Exams

Student	Mark	Laud	Course
276545	28	0	01
276545	27	0	04
937653	25	0	01
200768	30	1	04

```
TABLE Exams (  
    student integer,  
    mark integer,  
    course char(3),  
    laud bool,  
    CHECK (mark>=18 and mark<=30) ,  
    CHECK (not laud or mark=30) ) ;
```

Referential Constraints

Exams

Student	Mark	Laud	Course
276545	28	0	01
276545	27	0	04
937653	25	0	01
200768	30	1	04

```
TABLE Exams (  
    student integer,  
    mark integer,  
    course char(3),  
    laud bool,  
    CHECK (mark>=18 and mark<=30),  
    CHECK (not laud or mark=30),  
    FOREIGN KEY(student) REFERENCES  
        Students(studentID) ,  
    FOREIGN KEY(course) REFERENCES Courses(code) ) ;
```

Schema with Integrity Constraints

```
TABLE Students(  
    studentID integer Primary Key,  
    surname char(20),  
    name char(20),  
    DateofBirth date);
```

```
TABLE Exams(  
    student integer,  
    mark integer,  
    laud bool,  
    course char(3),  
    FOREIGN KEY(course) REFERENCES Courses(code),  
    FOREIGN KEY(student) REFERENCES Students(studentID),  
    UNIQUE(studentID, course));
```

```
TABLE Courses(  
    code char(3) Primary Key,  
    title char(50),  
    teacher char(20),  
);
```